

Model fit and parameter uncertainty in variational autoencoder based item response theory

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Abstract

Amortized variational inference (AVI) has become a computationally efficient alternative to marginal maximum likelihood estimation (MML) for estimating high-dimensional item response theory models on large datasets. However, currently the AVI framework lacks methods to assess model fit and determine parameter uncertainty. The challenge is that AVI optimizes a lower-bound to the marginal likelihood which does not guarantee the required asymptotic properties to obtain standard errors and the conventional model fit measures. In this paper, we use AVI estimates in a likelihood-based Fisher scoring procedure, ensuring asymptotic guarantees associated with MML. Large-sample theory for one-step estimators shows that when the initial estimator is sufficiently accurate (within an $O_p(1/\sqrt{N})$ neighborhood of the true parameter), a single Fisher scoring update is sufficient to obtain a consistent, and asymptotically efficient estimator. We investigate empirically whether one scoring step suffices to restore the validity of classical goodness-of-fit measures and standard error estimation. Using simulation studies and an application to large-scale Narcissism Personality Inventory data, we show that the proposed AVI-initialized one-step Fisher scoring approach yields fit statistics and uncertainty estimates that are on par with MML while retaining the computational advantage of AVI.

Keywords: amortized variational inference; item response theory; model fit; Fisher scoring; parameter uncertainty

Data and Code Availability Statement

All data used in both our simulation studies and our data application are available on OSF ¹. We provide all data generating parameters, generated datasets, parameter estimates and results. Additionally, all code required to implement our estimation routine is available on GitHub ².

1 Introduction

Item response theory (IRT) is a valuable tool for psychological and educational measurement, (Van der Linden, 2016). By modeling the relationship between

¹https://osf.io/xajcr/overview?view_only=0e12d919b05f4d669f595baa6760709b

²<https://anonymous.4open.science/r/GOF-PU-C88E/README.md>

latent psychological attributes and observed item responses, IRT enables researchers to construct valid measurement instruments (e.g. Kline, 2013), compare individuals on common latent scales (e.g. Kolen & Brennan, 2013; Wainer et al., 2000), and evaluate the psychometric quality of tests (e.g. Reise & Waller, 2009). In recent years, the scale of psychological measurement has increased substantially. High-dimensional models and large-scale datasets have become more common due to novel data collection techniques and large-scale assessment programs (Zenisky & Sireci, 2014; Thomas, 2019). In practice, this can lead to problems, as estimating high-dimensional IRT models is computationally challenging, especially on large datasets.

The standard marginal maximum likelihood estimation (MML) approach requires integrating over these high-dimensional variables (Reckase, 2009), which quickly becomes computationally prohibitive. Furthermore, the number of integrals to compute in each step grows linearly with the number of observations, amplifying the computational burden on large datasets. These limitations have motivated the adoption of variational inference as an alternative estimation framework. Rather than maximizing the marginal log-likelihood, which contains this high-dimensional integral, variational methods formulate an approximate posterior, which approximates the true posterior by optimizing a lower bound to the marginal log-likelihood (Blei et al., 2017). Such methods have been applied successfully to estimate multidimensional IRT parameters much more efficiently than MML (Cho et al., 2021; Ma et al., 2023). Although these approaches avoid the need for numerically expensive integrals, they still need to estimate a posterior for each participant, which can be costly in large scale applications. In Amortized variational inference (AVI), the parameters of this approximate posterior are not estimated directly, but rather by a neural network that is trained to predict the posterior parameters based on the observed response pattern. The parameters of this network are trained jointly with the model parameters using backpropagation (Curi et al., 2019). Burda et al. (2015) and Urban & Bauer (2021) further improve the approximation by taking multiple importance weighted samples from the approximate posterior. When the number of importance weighted samples goes to infinity, this lower-bound approaches the marginal log-likelihood. The authors show that using a small number of importance samples yields accurate parameter estimates in practice while being much faster than marginal MML, especially for high-dimensional models and large datasets. Further research has extended this idea to various different models and contexts (e.g. Liu et al., 2022; Paaben et al., 2022; Luo & Ji, 2025).

AVI has a big potential for efficient parameter estimation. However, parameter estimation is just one component of applied IRT analysis. In practice, IRT is typically used in combination with an extensive diagnostic toolbox for the evaluation of model fit, model comparison, quantification of parameter uncertainty, and the guidance of model refinement. These diagnostic procedures are central to assessing whether an estimated model provides a good representation of the data. However, currently the AVI framework lacks methods to assess model fit and determine parameter uncertainty. That is, popular fit statistics such the M_2 test, root mean square error of approximation (RMSEA), and

Comparative Fit Index (CFI) rely on asymptotic consistency of the estimator (Maydeu-Olivares & Joe, 2006; Browne & Cudeck, 1992; Bentler, 1990). Although previous research has shown that AVI parameter estimates are on par with MML estimates (Curi et al., 2019; Urban & Bauer, 2021; Veldkamp et al., 2025), the theoretical guarantees of MML only apply as the number of posterior samples goes to infinity, while in practice these are limited. As a result, AVI estimates cannot readily be used to obtain standard model diagnostics. That is, parameter consistency ensures that goodness-of-fit statistics converge to their theoretical distributions under the null hypothesis, and that estimated standard errors accurately reflect parameter uncertainty. Violations of this assumption might lead to incorrect judgements about model fit. Similarly, failing to meet these asymptotic properties complicates inference for the model parameters. Standard errors are usually obtained from the inverse of the Fisher information matrix, or from using the sandwich estimator. While the Fisher information requires correct model specification, the sandwich estimator remains asymptotically valid under misspecification (Van der Vaart, 2000; Huber et al., 1967). However, the resulting standard errors are valid only for the maximizing value of the estimating function, which is not necessarily the true parameter value, meaning that confidence intervals constructed using sandwich standard error might not have the intended coverage rate for the true parameter when the estimator is not consistent.

Therefore, in this study, we build on large-sample theory to derive a principled estimation strategy that attains both objectives. Specifically, we use the AVI parameter estimates in a Fisher-scoring procedure on the marginal log-likelihood, resulting in a consistent and asymptotically efficient estimator. Additionally, we emphasize a little utilized result in asymptotic statistics (Van der Vaart, 2000, Section 5.7) which holds that a single step of Fisher-scoring initialized at a sufficiently accurate preliminary estimator is enough to obtain a consistent and asymptotically efficient estimator with the same limiting distribution as the maximum likelihood estimator. Under regularity conditions which we elaborate in section 3, a single scoring update is therefore enough, and doing multiple steps does not affect the asymptotic distribution of the estimator. Such Fisher scoring steps are computationally expensive for high dimensional models, but by using AVI as preliminary estimator, we show that just a single step can be sufficient. As such, by using our proposed method we can obtain model fit diagnostics and standard errors, while still retaining the computational advantage of AVI over MML.

While in our proposed method we do not attempt to prove mathematically that this preliminary estimate is always sufficiently close for AVI estimates, in the study reported below we find that AVI estimates are close enough to the MML estimates to warrant relying on the single-step-estimator asymptotic theory. Therefore, one Fisher scoring step is adequate to achieve the desired asymptotic behavior in practice. We investigate whether classical goodness-of-fit measures such as M_2 , RMSEA, and CFI perform adequately after our one-step procedure. In addition, we investigate if the Fisher information matrix provides practically useful standard error estimates for the model parameters.

128 The rest of the paper is structured as follows: First, we formally introduce
 129 AVI, and show how it is used in estimating IRT parameters. Next, we intro-
 130 duce our estimation strategy, which aims to address the asymptotic issues with
 131 AVI parameter estimates. In the subsequent two sections, we evaluate the two
 132 important applications of this method alluded to above: goodness-of-fit and the
 133 estimation of parameter standard errors. Both sections include a simulation
 134 study which evaluates the performance of our proposed estimation procedure
 135 compared to standard MML and standard AVI. Finally, in section 6 we demon-
 136 strate the practical applicability of our approach on a large dataset of responses
 137 to a narcissism questionnaire, evaluating model fit and parameter uncertainty
 138 for high-dimensional MIRT models on large datasets, which would be practi-
 139 cally infeasible to fit using MML. We conclude with a discussion and several
 140 recommendations.

141 2 Amortized Variational Inference

142 AVI provides a computationally efficient alternative to MML estimation in latent
 143 variable models by avoiding repeated integration over the latent variables for
 144 each respondent. Here we briefly outline the approach, for a more detailed
 145 account see (Urban & Bauer, 2021).

146 Let $\mathbf{x}_i = (x_{ij})$ denote the binary response vector of a respondent i , $i =$
 147 $1, \dots, N$, on item j . Let ψ_j denote the vector of item parameters for item j and
 148 let $\boldsymbol{\psi} = (\psi_j)$ denote the collection of item parameters. Furthermore, let $\boldsymbol{\theta}$ denote
 149 a vector of latent variable scores in a Multidimensional item response theory
 150 (MIRT) model. In AVI, rather than maximizing the marginal log-likelihood

$$\ell(\boldsymbol{\psi}) = \sum_{i=1}^N \log \int p(\mathbf{x}_i | \boldsymbol{\theta}, \boldsymbol{\psi}) p(\boldsymbol{\theta}) d\boldsymbol{\theta}, \quad (1)$$

151 which is analytically intractable in multidimensional IRT, and computationally
 152 often infeasible, a variational inference approach is taken: AVI introduces an
 153 approximate posterior distribution $q_\phi(\boldsymbol{\theta}_i | \mathbf{x}_i)$ that is highly parameterized³ by a
 154 parameter vector ϕ , and optimizes a lower bound to the marginal log-likelihood.
 155 For each respondent i , the *evidence lower bound (ELBO)* is defined as

$$\text{ELBO}_i(\boldsymbol{\psi}, \phi) = \mathbb{E}_{q_\phi(\boldsymbol{\theta}_i | \mathbf{x}_i)} [\log p(\mathbf{x}_i | \boldsymbol{\theta}_i, \boldsymbol{\psi})] - \text{KL}(q_\phi(\boldsymbol{\theta}_i | \mathbf{x}_i) \| p(\boldsymbol{\theta}_i)), \quad (2)$$

156 and the parameters $(\boldsymbol{\psi}, \phi)$ are estimated jointly by maximizing $\text{ELBO}(\boldsymbol{\psi}, \phi) =$
 157 $\sum_{i=1}^N \text{ELBO}_i(\boldsymbol{\psi}, \phi)$. Here $\text{KL}(q(\cdot) \| p(\cdot))$ denotes the Kullback-Leibler divergence
 158 of density $q(\cdot)$ relative to $p(\cdot)$. As the name suggests, the ELBO is a lower bound
 159 on the marginal log-likelihood.

160 In practice, the tightness of the ELBO as a lower bound depends on the
 161 flexibility of the variational family $q_\phi(\boldsymbol{\theta}_i | \mathbf{x}_i)$. A highly factorized distribution
 162 is often used, which might result in a large gap between the ELBO and the

³usually by means of a neural network

163 true marginal log-likelihood. To address this issue, Burda et al. (2015) propose
 164 to use importance-weighted sampling with the approximate posterior as pro-
 165 posal distribution and the true joint distribution as the target. This yields the
 166 importance-weighted ELBO (IW-ELBO)

$$\text{IW-ELBO} = \mathbb{E}_{1:S} \left[\log \frac{1}{S} \sum_{s=1}^S \frac{p_{\psi}(\tilde{\boldsymbol{\theta}}^{(s)}, \mathbf{x})}{q_{\phi}(\tilde{\boldsymbol{\theta}}^{(s)} | \mathbf{x})} \right], \quad (3)$$

167 where S is the number of importance-weighted samples, and $\tilde{\boldsymbol{\theta}}^{(s)} \sim q_{\phi}(\boldsymbol{\theta} | \mathbf{x})$
 168 are samples drawn from the approximate posterior distribution. As $S \rightarrow \infty$,
 169 the IW-ELBO converges (in probability, pointwise) to the true marginal log-
 170 likelihood, provided $p_{\psi}(\boldsymbol{\theta}, \mathbf{x})/q_{\phi}(\boldsymbol{\theta} | \mathbf{x})$ is bounded.

171 In the context of multidimensional IRT, in simulation studies AVI has been
 172 shown to produce accurate item parameter estimates $\boldsymbol{\psi}$ while substantially re-
 173 ducing computation time compared to marginal maximum likelihood (Curi et
 174 al., 2019; Urban & Bauer, 2021). However, because AVI optimizes a lower
 175 bound rather than the marginal likelihood itself, the asymptotic distribution of
 176 the estimator is not guaranteed to be equal to that of the MML estimator.

177 3 The One-Step Estimator

178 The main limitation of amortized variational inference in IRT is the absence of
 179 asymptotic guarantees that make MML estimators so appealing. As a result,
 180 standard IRT tools for inference such as goodness-of-fit testing and the compu-
 181 tation of standard errors cannot be applied directly. Although increasing the
 182 number of importance-weighted samples ensures that the importance-weighted
 183 evidence lower bound (IW-ELBO) approaches the MML objective pointwise
 184 in the stochastic limit, this does not guarantee convergence of their estimates
 185 (Van der Vaart, 2000)⁴. Moreover, a finite value has to be chosen in practice,
 186 which needs to be computationally feasible. This may well result in an estimator
 187 that does not have the desired asymptotic properties of the MML-estimator.

188 We propose to address this issue by using the AVI estimates to initialize
 189 a Fisher scoring procedure. The Fisher scoring procedure ensures asymptotic
 190 consistency of the estimator, while the fast AVI estimator results in a very good
 191 initialization, allowing for a low number of Fisher scoring steps.

192 A little used insight from large sample theory is that a single Newton-
 193 Raphson step towards the optimum of the (marginal) likelihood initialized at a
 194 sufficiently accurate preliminary estimate $\tilde{\boldsymbol{\psi}}$ is enough to obtain an estimator
 195 that is consistent and asymptotically normal (Van der Vaart, 2000, Section 5.7).
 196 More specifically,

$$\hat{\boldsymbol{\psi}}_N = \tilde{\boldsymbol{\psi}}_N - [H_N(\tilde{\boldsymbol{\psi}}_N)]^{-1} H_N(\tilde{\boldsymbol{\psi}}_N), \quad (4)$$

⁴A sufficient condition would be *uniform* convergence in probability (Van der Vaart, 2000), but this is an unnecessarily strong assumption and we do not offer a proof for such strong convergence.

197 produces an estimator $\hat{\psi}_N$ with the asymptotic properties of the MML estima-
 198 tor, provided $\tilde{\psi}$ is within a $N^{-1/2}$ -range of the true value, Here $U_N = \frac{\partial \log p}{\partial \psi}$
 199 denotes the score function, $H_N = \frac{\partial}{\partial \psi^T} U$ denotes the Hessian matrix of the
 200 log-likelihood, and N is the sample size. Note that the requirements on the
 201 preliminary estimator are relatively weak. It is not required to maximize the
 202 log-likelihood; the only requirement is that, in formal terms,

$$\|\tilde{\psi}_N - \psi^*\| = O_p(N^{-1/2}), \quad (5)$$

203 where $\|\cdot\|$ denotes the Euclidean norm and $O_p(\cdot)$ denotes stochastic order. The
 204 main advantage of this approach is that it separates the problem of consis-
 205 tency from the problem of asymptotic efficiency. A preliminary estimator only
 206 needs to be sufficiently close to the true value, which can often be achieved
 207 using approximate methods. The subsequent one-step update then recovers the
 208 asymptotic distribution of the full maximum likelihood estimator. Importantly,
 209 asymptotic theory shows that a single Newton–Raphson step is sufficient to at-
 210 tain the same limiting distribution as the fully iterated estimator. Additional
 211 iterations can improve performance in a finite sample but do not yield asymp-
 212 totic improvement.

213 The intuition for this finding comes from the fact that, given a preliminary
 214 estimator that is sufficiently close to the true parameter ψ^* , the behavior of the
 215 score function is simple. Within a neighborhood of size $1/\sqrt{N}$ around the true
 216 parameter, the change in the score function when moving from a preliminary
 217 estimator $\tilde{\psi}$ can be approximated by a linear function. As the sample size
 218 grows without bound, the true change converges to the linear approximation in
 219 probability. Formally, Van der Vaart (2000) states the necessary condition

$$\sup_{\|\psi - \psi^*\| \leq M/\sqrt{N}} \left\| \sqrt{N}(U_N(\psi) - U_N(\psi^*)) - U^* \sqrt{N}(\psi - \psi^*) \right\| \xrightarrow{p} 0, \quad (6)$$

220 for some fixed constant $M > 0$. Here U_N in our case is the score, ψ^* denotes
 221 the true parameter and U^* is a nonsingular matrix giving the local linear ap-
 222 proximation of the score around the true parameter value. In fact, U^* can be
 223 replaced by a sequence of matrices $U_{N,0}$ that converges to U^* in probability.

224 We propose to use the AVI estimate as the preliminary estimator and im-
 225 prove it with a single Fisher scoring step for maximizing the marginal likelihood
 226 function of the item response theory model in order to obtain an estimator with
 227 the desirable asymptotic properties of MML. That is, we replace the hessian
 228 $[H_N(\tilde{\psi}_N)]^{-1}$ in the Newton Raphson algorithm above by the outer product of
 229 gradients estimator of the Fisher information:

$$I_N(\psi) = \frac{1}{N} \sum_{i=1}^N U_i(\psi) U_i(\psi)^\top, \quad (7)$$

230 where $U_i(\psi)$ denotes the individual score contribution, defined as the gradient of
 231 the individual marginal log-likelihood with respect to ψ . Note that computing

the score requires high-dimensional integration over the latent variables, which can be computed using numerical quadrature (R. D. Bock & Aitkin, 1981). Specifically, we use Gauss-Hermite quadrature to approximate the score. This numerical integration is computationally demanding. However, since asymptotic theory requires only a single update, it still offers a substantial computational advantage over MML methods, which must perform the same integration at every iteration.. The outer product of gradients estimator then gives the update step:

$$\hat{\psi}_N = \tilde{\psi}_N + I_N(\tilde{\psi})^{-1} U_N(\tilde{\psi}_N), \quad (8)$$

which is computationally feasible, even for high dimensional models.

An important consideration is whether the AVI estimator converges to the true value ψ^* when the number of importance-weighted samples is finite. While we know that theoretically an infinite number of samples is needed for the ELBO to approach the log likelihood, the requirement imposed by the one-step estimator does not require full convergence to the MML estimator. The preliminary estimator only needs to lie within a $O_p(N^{-1/2})$ neighborhood of the true parameter value. Other theoretical guarantees, such as likelihood optimality or asymptotic normality are not required. Although we do not derive this consistency directly for a finite number of samples, existing simulation studies consistently show empirically that importance-weighted AVI with a relatively small number of samples already produces parameter estimates that are as close to the true parameter values as MML estimates (Urban & Bauer, 2021; Liu et al., 2022). Our simulations will investigate further whether these preliminary AVI estimates are sufficient for one-step inference.

Since the estimates obtained from this procedure are expected to be consistent, we will use these to compute model fit statistics. In addition, the estimates obtained using the one-step procedure can readily be used to obtain asymptotically correct standard errors. We will describe the exact methods used for evaluating goodness-of-fit and parameter uncertainty in the corresponding sections.

4 Goodness-of-fit in IRT

Evaluating goodness-of-fit is essential in latent variable modeling. Before interpreting latent traits or making practical decisions, it is important to know whether the model reproduces key features of the data. For dichotomous data, traditional tests of goodness-of-fit are generally based on χ^2 tests comparing the observed frequencies of each response pattern and the expected frequencies of each response pattern under the model (e.g. D. Bock & Lieberman, 1970). For dichotomous data with J items, there are 2^J possible response patterns, and the Pearson chi-square goodness-of-fit statistic is

$$\chi^2 = \sum_{r=1}^{2^J} \frac{(O_r - E_r)^2}{E_r}, \quad (9)$$

where O_r denotes the observed frequency of pattern r , and $E_r = Np_r(\hat{\theta})$ is the model implied expected frequency. Although this test works well in theory, a well-known fact about the χ^2 test statistic is that it deviates from the theoretical distribution when many cells have expected frequencies of less than 5 (Cochran, 1954). For dichotomous data, the number of possible response patterns grows exponentially with the number of items, which inevitably leads to small expected cell counts as the number of items grows too large. Maydeu-Olivares & Joe (2006) introduced a new test statistic which does not suffer from this problem. Rather than comparing observed and expected counts of full response patterns, the test compares observed and model-expected moments of individual items and low-order item combinations. They define the general limited-information M statistic as

$$M = N[\hat{\mathbf{p}} - \mathbf{p}(\hat{\psi})]^\top W[\hat{\mathbf{p}} - \mathbf{p}(\hat{\psi})], \quad (10)$$

where $\hat{\mathbf{p}}$ is the vector of observed proportions for the set of marginal probabilities included in the analysis (e.g., item proportions, bivariate proportions, trivariate proportions, etc.). Let $\mathbf{p}(\psi)$ be the corresponding vector of model-implied proportions, and let W be a suitable weight matrix (often the inverse of the asymptotic covariance matrix of $\hat{\mathbf{p}}$). A commonly used special case is the M_2 statistic, obtained when $\hat{\mathbf{p}}$ consists of all univariate proportions $\hat{p}_j$ and all bivariate proportions $\hat{p}_{jk}$ for $j < k$. Under standard regularity conditions, and provided that $\hat{\mathbf{p}}$ is a consistent and asymptotically normal estimator of the true marginal proportions, this test statistic is asymptotically χ^2 distributed, with $df = S - k$, where S is the total number of marginal proportions used and k is the number of model parameters.

Where significance tests such as the Pearson χ^2 test or the limited-information M_2 test can provide insight into the discrepancy between the model predictions and the observations, it is well-known that even models that approximate the data well may still be rejected as unimportant discrepancies between the model and the data may become significant if the sample size is large enough (Browne & Cudeck, 1992). For that reason, it is often more useful to use indices of approximate fit like the RMSEA (Browne & Cudeck, 1992). The RMSEA aims to summarize the approximation error, that is, the discrepancy between the model implied population distribution and the actual population distribution. The RMSEA can be determined from the Pearson chi-square above, that is,

$$\text{RMSEA} = \sqrt{\frac{\max(\chi^2 - df, 0)}{df(N - 1)}}, \quad (11)$$

which is a measure of the approximation error per degree of freedom, thereby accounting for model complexity.

Instead of approximate fit, a second common choice that avoids the problems related to significance testing is to focus on incremental fit, using for instance, the CFI (Bentler, 1990). The CFI is given by

$$\text{CFI} = 1 - \frac{\max(\chi_{\text{model}}^2 - df_{\text{model}}, 0)}{\max(\chi_{\text{base}}^2 - df_{\text{base}}, 0)}, \quad (12)$$

where χ_{base}^2 denotes the Pearson chi-square statistic of a baseline model where all items are uncorrelated and only the item means and variances are estimated. This metric measures the degree of improvement of the model over an independence model. In IRT applications, M_2 is often used in place of Pearson's χ^2 when computing approximate or incremental fit indices (Maydeu-Olivares & Joe, 2014; Cai et al., 2023).

In addition to these fit indices, another common tool used in the evaluation of model-fit are Lagrange multiplier (LM) tests. Where the fit indices discussed above focus on the global fit of the model, LM tests focus on local areas of misfit. Specifically, the LM statistic indicates for the constrained parameters in a model what the expected local increase in log-likelihood is if the constrained parameters will be freely estimated. As such, to evaluate parameter constraints, LM statistics can be considered even without actually estimating the unconstrained model (Aitchison & Silvey, 1958). The LM test statistic is given by

$$LM = U_N(\hat{\psi}_0)^\top I(\hat{\psi}_0)^{-1} U_N(\hat{\psi}_0), \quad (13)$$

where $\hat{\psi}_0$ denotes the full set of item parameters subject to the constraints. Under the conditions specified in Aitchison & Silvey (1958), the LM statistic follows a χ^2 distribution where the degrees of freedom is the number of constrained parameters. In the SEM context, the LM statistic is commonly referred to as a Modification Index (MI). The MI for a constrained parameter is simply the LM statistic associated with freeing that parameter (e.g. Bollen, 1989, Chapter 8). Consequently, a large MI indicates that releasing the constraint is expected to yield an improvement in model fit, without re-estimating the less constrained model. An important sidenote is that post-hoc model modifications are known to be susceptible to chance capitalization (MacCallum et al., 1992). Therefore, MIs are often used as diagnostic tools in practice, rather than direct tests of whether a parameter should be freed. Although the theoretical $\chi^2(df = 1)$ implies a critical value of 3.84 at the significance level $\alpha = .05$, popular statistical software only reports MIs above 10 in order to prevent such chance capitalisation (Muthén & Muthén, 2017).

Of course all three sets of tools discussed above (i.e. tests of absolute fit, approximate and incremental fit indices, and MIs) rely on the assumption that the estimator is consistent. As discussed in the introduction, this limits their applicability when parameters are estimated using AVI, as this assumption is not necessarily guaranteed in the AVI setting. We propose that by using the one-step procedure outlined in section 3, we can ensure these asymptotic guarantees, allowing for the use of traditional fit measures. To demonstrate that this procedure can indeed result in accurate goodness-of-fit statistics, we perform a simulation study where we estimate models under various levels of misspecification and evaluate the type-I and type-II error rates of the AVI-one-step procedure, compared to both AVI without the one-step procedure and to standard MML.

349 4.1 Simulation 1: Goodness of Fit

350 In the first simulation study we evaluate the performance of goodness-of-fit tests
351 of our proposed procedure under different levels of misspecification

352 We achieve different levels of misspecification by varying the number of ad-
353 ditional cross-loadings that are not estimated in the model (from 0 to 9), and by
354 varying the size of these cross loadings (small, medium or large). We simulate
355 100 datasets for each condition, resulting in 3000 datasets. Additionally, on each
356 dataset we are evaluating 3 methods (MML, AVI and the one-step approach)
357 resulting in a $10 \times 3 \times 3$ factorial design, with 100 replications for each cell. Note
358 that in this fully crossed design, there are actually three conditions which are
359 the same, since the size of the cross loadings is irrelevant when the number of
360 cross loadings is zero. We decided to keep all three conditions, resulting in 300
361 datasets without any misspecification. These additional replications increase
362 the precision of our type-I error estimates.

363 Data is generated from three-dimensional IRT models. In all conditions an
364 IRT model is estimated with the same factor structure. However, as discussed
365 above, additional cross loadings of various sizes are introduced during simulation
366 in order to manipulate the degree of misspecification. The true intercepts are
367 drawn from a Uniform distribution from -2 to 2, and true abilities are drawn from
368 a multivariate standard Normal distribution. The base set of slopes, which is
369 shared across all datasets, is drawn from a Uniform distribution between .5 and
370 2. Depending on the condition, several cross loadings are added to this base set,
371 resulting in a misspecified model. Specifically, the number of additional cross
372 loadings that are not estimated in the model varies between zero and nine, and
373 the size of these cross loadings is manipulated into three levels: a small, medium
374 or large misspecification. In the small condition, cross loadings are sampled from
375 a $\mathcal{U}(.1, .2)$, in the medium condition from $\mathcal{U}(.2, .3)$ and in the large condition
376 from $\mathcal{U}(.3, .4)$. The exact parameters used in each condition are available in the
377 accompanying Github repo.

378 We compare rejections rates for the three methods across different levels of
379 misspecification, investigating whether the type-II error rate of the proposed
380 AVI and one-step procedure are up to par with the type-II error rate based on
381 traditional MML estimates. MML estimation is done with the EM algorithm
382 implemented in the *mirt* R package, using the default settings (Chalmers, 2012).
383 AVI estimation is implemented in PyTorch (Paszke et al., 2019). We use 25
384 importance weights, and set the number of nodes in the hidden layer to 16 ($(J$
385 $+ 2D)/2$), we use a learning rate of .005, the setting of these hyperparameters
386 is based on recommendations by (Urban & Bauer, 2021). In addition to looking
387 at the type-II error rate, we also investigate the distribution of the test statistic
388 under a correctly specified model, as well as the type-I error rate. Rather than
389 doing this in a separate simulation study, we simply take the conditions in which
390 there are no additional cross loadings. As mentioned previously, there are 300
391 datasets without misspecification, due to the fully crossed design.

392 One thing to note is that in this simulation, we assume a single set of primary
393 loadings, and can therefore not provide insight into the effect of different data

Table 1: Results when model is correctly specified (N=600) We report the percentage of times the null was rejected, the Kolmogorov-Smirnov distance to the theoretical χ^2 distribution (df=253) and the associated p-value, testing the null hypothesis that the distribution of M_2 test statistics matches the theoretical χ^2 .

Method	Reject Rate	KS distance	p-value
AVI	3%	.081	.039
One-step	4.33%	.035	.860
MML	4.33%	.036	.835

generating parameters on the behavior of the M_2 statistic. However, we provide a supplementary simulation study in the supplementary material, demonstrating that our one-step approach results in the same behavior of the tests based on MML estimates.

5

4.1.1 Results

Figure 1 presents the distribution of the M_2 test statistic under correct model specification for the three different methods, as well as the theoretically implied χ^2 density. Under correct specification, M_2 computed from MML closely follows the reference χ^2 distribution. In contrast, M_2 based on AVI is slightly shifted downward, suggesting a light distortion in the implied null distribution. After the one-step update, the M_2 distribution aligns closely with the reference and is nearly indistinguishable from MML. In addition to this visual confirmation that the one-step procedure is working, table 1 presents the rejection rates of the different methods as well as the Kolmogorov-Smirnov (KS) distance to the theoretically implied distribution. The results show that the rejection rate of AVI is too low, whereas the rejection rate of the one-step procedure exactly matches the MML approach. The KS distance of the AVI test statistic distribution to the implied distribution is also substantial, whereas the data do not suggest that the one-step or MML test statistics deviate from the implied distribution. These results reflect that the one-step procedure works as intended.

Figure 2 presents the proportion of times the null hypothesis was rejected for the varying number of cross-loadings and effect sizes. Results indicate that the probability of rejection grows with the number of cross-loadings and the effect size for all three methods, which is of course in line with expectations. The proportion of rejections is slightly smaller for AVI than for MML in most conditions, indicating that the goodness-of-fit test has slightly lower power when using AVI. However, the rejection rate of the one-step procedure matches the MML rejection rate, indicating that our one-step procedure behaves as theoretically expected.

⁵<https://anonymous.4open.science/r/GOF-PU-C88E/README.md>

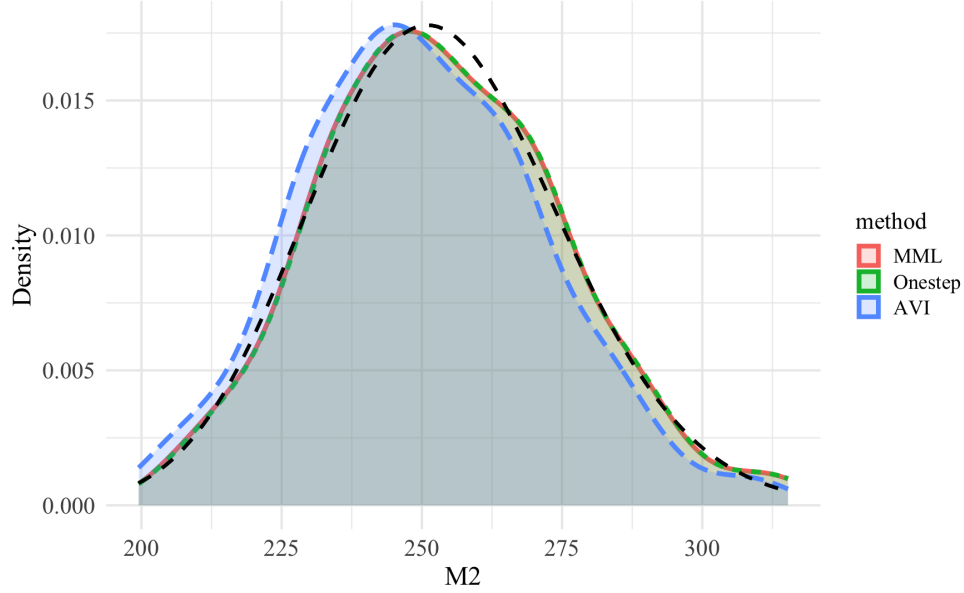


Figure 1: Density of the M_2 statistic for correctly specified models estimated using MML, AVI and one-step estimation. The theoretical χ^2 (df=253) is plotted in black. The number of replication used for this plot is 300.

5 Parameter uncertainty

A second advantage of the asymptotic guarantees that our one-step procedure provides is the ability of valid expression of parameter uncertainty. In an IRT context standard errors are generally obtained from the asymptotic covariance matrix of the item parameters. When parameters are estimated using MML, this can be computed as

$$\widehat{\text{Var}}(\hat{\psi}) = I(\hat{\psi})^{-1}, \quad (14)$$

where $I(\hat{\psi})$ is the empirical outer product of the score matrix expressed in equation 7. While this approach is valid under a correctly specified model, in practice there might be minor model misspecifications or local dependencies. An alternative method that is more robust to model misspecification is the sandwich estimator proposed by (Huber et al., 1967). Rather than relying solely on the Fisher information matrix, this approach combines the expected Fisher information matrix with the inverse of the negative Hessian of the log-likelihood

$$\widehat{\text{Var}}(\hat{\psi}) = H_N^{-1} I(\hat{\psi}) H_N^{-1}. \quad (15)$$

Recall that H_N denotes the Hessian of the log-likelihood. When the model is correctly specified, it holds that at the maximum of the likelihood function: $I(\hat{\psi}) = -H(\hat{\psi})$, which means the first two terms cancel, and we are left with

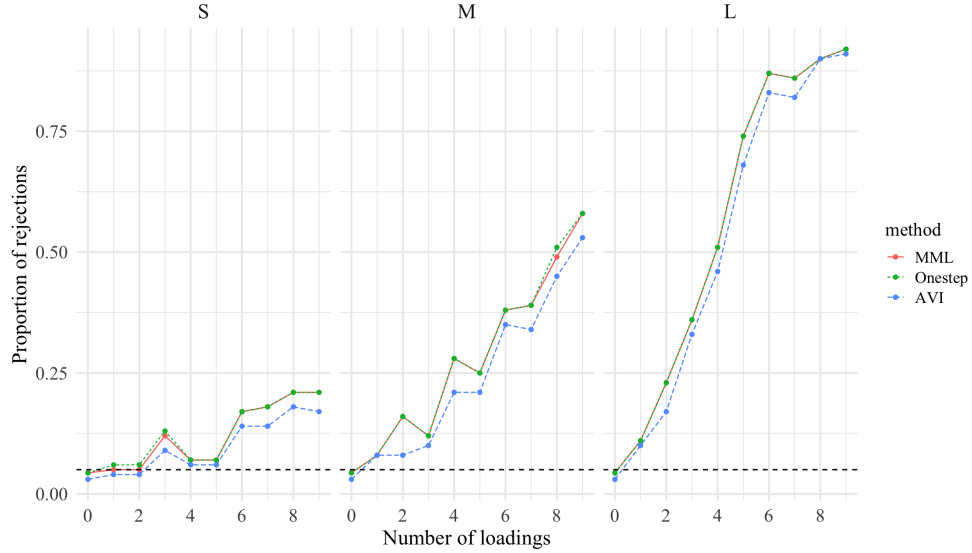


Figure 2: Proportion of times the null hypothesis is rejected for varying degrees of misspecification for models estimated using MML, AVI and one-step estimation. The three panels represent small, medium and large cross loadings and the x axis represent the amount of misspecified cross loadings. The horizontal black dotted line represent the 0.05 α level.

the original expression from equation 14. However, when the model is slightly misspecified, this estimator can provide valid standard errors by correcting for this model misspecification.

Although both aforementioned methods are valid estimators of the asymptotic covariance matrix, they both rely on the assumptions of consistency. As mentioned previously, these guarantees are not necessarily ensured for AVI estimates, meaning that using these estimators directly on AVI estimates is not theoretically justified.

A third common approach for estimating standard error is bootstrapping. Rather than relying on asymptotic behavior of the estimator, this approach estimates parameter uncertainty by repeatedly resampling data and reestimating parameters. This provides a non-parametric approximation to the parameter sampling distribution, with the advantage of not relying on strong distributional assumptions. However, it comes at the cost of computational efficiency, as it requires reestimating the parameters many times.

In the current section, we investigate the behavior of these standard error estimates for AVI method both before and after the proposed one-step procedure.

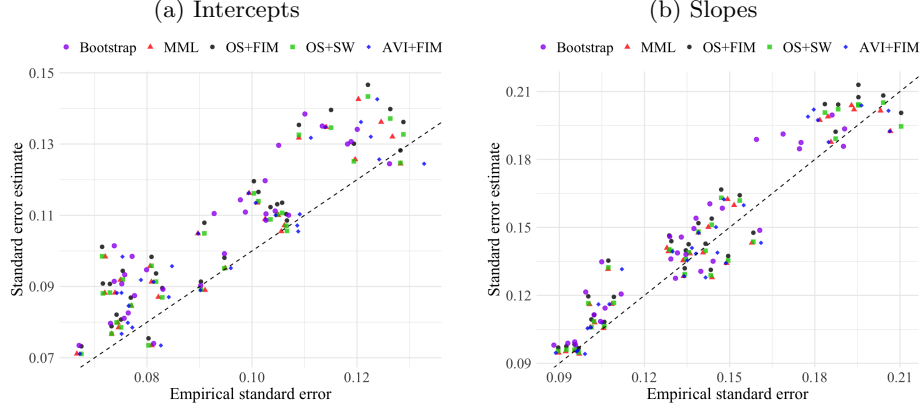


Figure 3: Standard error estimates for intercepts (3a) and slopes (3b) plotted against the empirical standard error over 100 iterations. Standard error estimates either using the Fisher information matrix directly after applying the one-step procedure, using the sandwich estimators after applying the one-step procedure, using bootstrapping without the one-step procedure or using Oakes method based on the MML estimates .

5.1 Simulation 2: Parameter Uncertainty

In order to evaluate the standard errors produced by the different methods, we simulate 100 datasets from a MIRT model. We use a three-dimensional model with simple structure (the exact structure is available on GitHub), and simulate datasets of 1000 observations. In this simulation study, we estimate MIRT parameters using the correctly specified factor structure. For each of the datasets, we estimate parameters by using our proposed AVI and one-step procedure. We produce estimates of the standard errors using the three discussed methods. Namely by taking the inverse of the expected Fisher information matrix, by using the sandwich estimator, or by performing bootstrap estimation of the standard errors without performing the one-step procedure. For the sake of comparison, we also estimate the parameters using MML using `mirt`, and estimate standard errors. These standard errors are computed using Oakes method, a numerical approximation of the observed information matrix (Chalmers, 2018). This is the default approach for computing standard errors in `mirt`.

Table 2: Standard error estimates across methods (N=1000, 100 replications). We report the coverage rate of the 95% confidence interval and the average correlation between the estimated standard errors and the empirical standard deviation across replications. AVI and FIM use the Fisher information matrix applied to AVI estimates directly and after the one-step update respectively; SW uses the sandwich estimator after the one-step update; MML uses Oakes’ method on MML estimates; Bootstrap resamples from AVI estimates without the one-step update.

Method	Intercepts		Slopes	
	Coverage	Corr.	Coverage	Corr.
AVI	.948	.905	.946	.954
FIM	.952	.913	.960	.960
SW	.948	.912	.948	.961
MML	.947	.914	.955	.960
Bootstrap	.942	.897	.955	.946

Figure 3a shows the estimated standard errors of the item slopes for the four different standard error methods, and figure 3b shows the same for the intercepts. Overall, the figure shows that the estimated standard errors closely match the empirical standard error across replications. It is important to note that although the different methods perform equally in this specific setting, the one-step procedure is still valuable because of the asymptotic guarantees that it provides. Table 2 shows that all methods yield coverage rates close to the nominal 95%. To assess whether coverage probabilities differed across methods, separate chi-square tests of independence were conducted for intercepts and slopes, based on contingency tables of method by coverage (covered vs. not covered). The tests were not statistically significant both for intercepts $\chi^2(4) = 3.08$, $p = .54$, and slopes $\chi^2(4) = 8.73$, $p = .068$, indicating that no differences among methods were detected. These results suggest that all five methods provide adequate confidence intervals in the current setting, even when using AVI without our proposed one step procedure.

6 Application

To demonstrate the practical utility of our approach we apply our method to the Narcissism Personality Inventory (NPI), which is a psychological inventory measuring narcissistic personality traits developed by R. N. Raskin & Hall (1979). The original questionnaire consists of 54 forced-choice binary questions where one option is considered as narcissistic and the other is not. However, most studies, including our analysis, consider only a 40 item subset of the inventory.

The dimensionality of the narcissism construct has been debated, since different studies have recovered different factor structures. Originally, Emmons (1984) analysed responses to the 54 item NPI by 451 undergraduate

498 students. They used principal component analysis on the Pearson correla-
499 tion matrix of the binary responses, retaining factors with eigenvalues larger
500 than one. They identified four components which were interpreted as leader-
501 ship/authority, self-absorption/self-admiration, superiority/arrogance and ex-
502 ploitativeness/entitlement.

503 Several years later, R. Raskin & Terry (1988) did a similar analysis on the
504 same version of the NPI on a sample of 1018 undergraduates. They dropped
505 7 items with poor psychometric attributes resulting in just 47 items that were
506 analyzed. They perform principal component analysis (PCA) on the tetrachoric
507 correlation matrix, to better account for the binary nature of the responses. The
508 authors use a multistep procedure for selecting the number of factors. They
509 started with an eight factor structure as these were the eight behavioral dimen-
510 sions listed in the DSM-III, but ultimately retained only seven. Their criteria for
511 retaining a factor were that each factor should have had at least three items with
512 loadings $> .5$ without a strong cross loading and that there are no substantial
513 negative item loadings. They labelled the seven factors as authority, exhibition-
514 ism, superiority, entitlement, exploitativeness, self-sufficiency and vanity. In
515 addition to this more theory based factor structure, another contribution was
516 that they show that a 40 item version of the questionnaire shows a very high to-
517 tal score correlation with the original scale (.98), leading subsequent researches
518 to only administer this more concise 40 item inventory.

519 More recently, Kubarych et al. (2004) used exploratory PCA on the 40 item
520 NPI in a sample of 338 Scottish undergraduates. Rather than starting from
521 a predetermined number of factors, the authors perform PCA on the Pearson
522 correlation matrix, considering solutions with increasing numbers of factors. In
523 addition to the exploratory PCA they also use Confirmatory factor analysis
524 (CFA) to fit models of different factor structures, and use Lagrange multiplier-
525 and Wald-tests to further tweak their model. Based on the CFA results they
526 found evidence for either a two-factor (Power and Exhibitionism) or three-factor
527 solution (adding a third weaker factor identified as 'being a special person').

528 In yet another study, Corry et al. (2008) analyse the 40 item NPI on a sample
529 of 1448 undergraduate students. They use an initial random subset of 724 college
530 students, for Exploratory factor analysis (EFA) on the tetrachoric correlation
531 matrix, which suggests a two factor solution labelled as leadership/authority
532 and exhibitionism/entitlement. They use the second subset of the other 724 stu-
533 dents to perform a comprehensive CFA, comparing the proposed factor struc-
534 ture to the four factor structures proposed by the 3 aforementioned studies.
535 Although fit indices support the 7 factor structure from (R. Raskin & Terry,
536 1988), they note that only one of these seven factor scales produced satisfac-
537 tory internal consistency. Finally they conclude that the two factor solutions
538 proposed by (Kubarych et al., 2004) and themselves provide the best balance
539 between goodness-of-fit and parsimony.

540 Finally, most recently Ackerman et al. (2011), have done another com-
541 prehensive comparison of the different structures on a much larger sample of
542 19001 college students. They perform a new EFA pointing to a three fac-
543 tor model including leadership/authority, grandiose exhibitionism and entitle-

ment/exploitativeness. They argue that Corry et al. (2008) under-extracts factors due to the focus on internal consistency. Additionally, they contribute by investigating the external validity of the different factors.

Where these factor structures have previously evaluated in a factor analytic framework, the current study will estimate multidimensional IRT models to model NPI data. Using the AVI tools discussed in our study, it is straightforward to estimate multidimensional IRT models, even with as many as seven dimensions, which would be very computationally challenging using MML. Specifically, to demonstrate the applicability of our proposed AVI approach on real data, we will apply it to 11,243 responses to the 40-item NPI obtained from the Open-Source Psychometrics Project. (Open-Source Psychometrics Project, 2012). We will estimate simple structure multidimensional IRT models based on the six factor structures proposed in the five aforementioned papers. A table of the exact factor structures used is available on github, along with the code necessary to reproduce our results. We estimate all IRT models in PyTorch (Paszke et al., 2019) using the same settings discussed in section 4.1.

The seven-dimensional IRT model was estimated using AVI in just 72 seconds on the CPU of a MacBook Air (2025). Performing the one step procedure using five quadrature points per dimension required an additional 8.2 minutes. Note that estimating the same model using MML would be extremely computationally intensive due to the high-dimensionality and the large sample size. Table 3 presents the results for this application. All p -values for the tests of absolute fit were significant, indicating that the data never matches the model implied distributions exactly. This is to be expected in such a large sample. The model fit indices for the CFA models indicate that the model using the seven-factor structure fits the model best, whereas the other models are quite close in terms of model fit, with the two indices sometimes disagreeing on the order of model fit. Although we focus exclusively on the one-step procedure here, goodness-of-fit tests based directly on AVI estimates led to substantially similar conclusions in this specific case. The value of the one-step procedure lies in the asymptotic guarantees that make formal interpretation of these fit statistics theoretically justified. The factor structures 3 factors by (Ackerman et al., 2011) and the 4 factor structure by (Emmons, 1984) have many high MIs. The other methods have relatively little high MIs. The MIs for the 7 dimensional model by R. N. Raskin & Hall (1979), which fits the data best, suggest that three items might load on an additional factor. Specifically, they suggest that item 1 ('I have a natural talent for influencing people') might also load on Exploitativeness in addition to Authority, item 14 ('I insist upon getting the respect that is due to me') might also load on Vanity instead of just Entitlement and item 40 ('I am an extraordinary person') might also load on Authority instead of just Superiority. However, it is important to remember that modification indices are data-driven and primarily serve as diagnostic tools. Freeing parameters solely on their MIs risks capitalizing on chance and should be guided by substantive theory.

Table 3: Results for the six different factor structures reported in the five different studies. The table respectively reports the study, the number of latent dimensions, the number of items retained in the proposed factor structure, M_2 statistic, its degrees of freedom, the p-value, the RMSEA, the CFI, the percentage of constrained item slopes with a MI larger than 10 and the maximum MI across all slopes.

Study	Dim.	I	$M_2(df)$	p	RMSEA	CFI	MI>10	max MI
Emmons	4	31	22163(425)	<.001	.067	.921	34.4%	90.37
Raskin	7	40	18084(713)	<.001	.047	.957	0.4%	17.22
Kubarych	3	37	21438(624)	<.001	.054	.943	6.8%	38.62
Kubarych	2	37	21832(627)	<.001	.055	.942	0%	5.91
Corry	2	23	13721(228)	<.001	.073	.908	8.7%	30.70
Ackerman	3	25	13748(270)	<.001	.067	.926	36%	112.08

7 Discussion

In this study we propose to use one step of Fisher scoring in combination with AVI in order to obtain model fit and parameter uncertainty measures for multidimensional IRT models. This allows us to use various tools from the IRT toolbox for evaluating model fit and parameter uncertainty. We show that goodness-of-fit test statistics obtained using our approach follow the theoretical distribution under the null hypothesis, and that the type-II error rate using our approach is up to par with goodness-of-fit tests based on models estimated using MML. Additionally, we show that standard errors obtained using our approach capture parameter uncertainty to the same extent as those obtained using MML. Finally we apply this approach to a large dataset of responses to the NPI inventory. Our approach allows us to estimate IRT models efficiently with up to seven latent dimensions, which would be infeasible using quadrature based MML.

An important assumption underlying the one-step estimator is that the preliminary estimator is within range $N^{-1/2}$ of the true value. In the current study, AVI is used as such a preliminary estimator. While it is known that importance-weighted AVI converges to the MML estimator as the number of importance samples goes to infinity, this theoretical result does not apply in practice, where the number of samples is finite. Importantly, however, the requirement imposed by the one-step estimator is considerably weaker than the assumptions needed to apply goodness-of-fit test and likelihood based standard errors directly, without using the one-step estimator. The AVI estimates do not need to maximize the likelihood function, or be asymptotically normal. The only requirement is that the estimator lies within a sufficiently small neighborhood around the true parameter value. Although we currently can not derive theoretically that AVI estimates using a finite number of posterior samples is always within range $N^{-1/2}$ of the true value, empirical results from both previous studies and the simulations presented here consistently show that AVI param-

eter estimates are sufficiently close to the corresponding MML estimates, even when using a relatively small number of importance samples. These findings suggest that in typical applications, the remaining discrepancy between AVI estimates and the true parameter values is small enough to satisfy the requirements of the one-step procedure. Additionally, consistency of the estimator can also be assured by taking multiple Fisher scoring steps (essentially doing regular MML with AVI as starting values), although this of course comes at the cost of a higher computational burden. Formally establishing conditions under which amortized variational estimators are within range $N^{-1/2}$ of the true value remains an important topic for future research.

Another important consideration regarding our findings with respect to goodness-of-fit testing is the dependence of goodness-of-fit tests on the data generating parameters. Recent literature on dynamic fit indices (McNeish & Wolf, 2023; McNeish, 2023) shows that the behavior of model fit indices such as χ^2 , RMSEA and CFI vary based on factors like the number of items and the magnitude of the loadings, and overall factor reliability. Although the results of our simulation study show that the behavior of goodness-of-fit tests following our procedure is up to par with MML for varying levels of misspecification, we did not investigate the behavior of the M_2 test under different data-generating models. Although a comprehensive investigation of all different characteristics on the behavior of the M_2 test is beyond the scope of the current paper, as discussed above, we did conduct a small supplementary simulation in the appendix in the supplementary material. to show that the behavior of the M_2 test based on our AVI and one-step approach is the same as the behavior of the M_2 test based on MML for different magnitudes of the primary loadings and for different numbers of items per factor. So although fit measures are indeed sensitive to the data generating model, this effect is the same with our one-step procedure as for MML, reflecting that this is a general challenge, and not specific to our approach.

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756 A Primary loadings in the goodness-of-fit sim- 757 ulation

758 As discussed in the main text, our goodness-of-fit simulation uses a three-
759 dimensional model where each item loads on one or two dimensions. The slope
760 parameters are drawn from $\mathcal{U}(.5, 2)$ and the intercepts are drawn from $\mathcal{U}(-2, 2)$.
761 The exact parameter values used are available in table 4. Please note that these
762 are only the set of primary loadings. Depending on the condition, additional
763 cross-loadings are added, which are not estimated by the model. We refer to
764 GitHub for the complete set of parameters in each condition ⁶.

⁶<https://anonymous.4open.science/r/GOF-PU-C88E/README.md>

Table 4: Base parameter values used for goodness-of-fit simulation. Additional cross-loadings were introduced depending on the experimental condition.

Intercept	Slope 1	Slope 2	Slope 3
0.136	0.938	-	-
0.742	1.179	-	-
1.968	1.196	-	-
-1.456	0.710	-	-
-0.149	1.799	1.083	-
-0.672	1.877	1.064	-
1.092	0.790	1.153	-
-0.159	1.435	1.508	-
-0.285	1.382	1.496	-
-1.266	0.720	1.337	1.694
0.510	0.600	0.769	1.765
0.100	1.550	0.844	1.895
0.855	1.395	1.443	1.902
0.815	0.734	1.566	0.973
-0.884	0.675	0.734	1.510
0.882	-	0.781	1.494
1.896	-	1.239	1.351
1.757	-	1.080	1.096
0.626	-	1.113	1.136
-1.709	-	1.994	1.914
1.334	-	-	1.913
-0.797	-	-	1.597
-1.520	-	-	1.819
-1.780	-	-	0.734
0.136	-	-	1.596

765 B Supplementary goodness-of-fit simulation

766 The results of our original goodness-of-fit simulation are based on a specific set of
767 model parameters that we considered representative for a multidimensional IRT
768 context. We used 25 items and sampled loadings from $\mathcal{U}(.5, 2)$. Importantly,
769 however, previous research has shown that the behavior of goodness-of-fit indices
770 such as M_2 is not universal, but depends on the data-generating parameters. In
771 particular, the factor reliability of a model is known to affect the interpretation
772 of fit indices, which in turn depends on the magnitude of the primary loadings
773 and the number of items per dimension. The aim of the current study is
774 to show that the one-step procedure produces the same asymptotic behavior as
775 MML, and a comprehensive analysis of the behavior of the M_2 under different
776 data generating distributions is beyond the scope of this paper. Nevertheless, we
777 provide a small supplementary simulation in which we vary the number of items
778 and the magnitude of the primary loadings to show that the observed behavior
779 is not specific to the model used in our main simulation. Specifically, in
780 this supplementary simulation, we generate data from a three-dimensional IRT
781 model with simple structure. We vary the number of misspecified cross-loadings
782 to either zero or five, in order to include cases where the model is correctly
783 specified as well as cases where the model is incorrectly specified. Additionally
784 we vary the number of items per dimension to either five or ten, resulting in
785 models with 15 and 30 items respectively. Finally, we vary the size of the primary
786 loadings to be high $\mathcal{U}(1.5, 2.5)$ or low $\mathcal{U}(.25, 1.25)$ resulting in a 2x2x2 factorial
787 design. We simulate 100 datasets for each condition and perform M_2 tests using
MML, AVI, and our one-step procedure. Table 5 contains the rejection rates

Table 5: Mean Rejection Rates by Loading Level, Method, Number of Loadings,
and Items per Factor

Items/Factor	cross-loadings	High Primary loadings			Low Primary loadings		
		AVI	OS	MML	AVI	OS	MML
5	0	.03	.03	.03	.03	.03	.03
	5	.55	.55	.55	.55	.54	.54
10	0	.06	.06	.06	.06	.06	.06
	5	.22	.22	.22	.25	.25	.25

788 in each of the 8 conditions and for each of the three estimation methods. The
789 results confirm that factor reliability plays an important part in the behavior of
790 the M_2 test. Although the size of the primary loadings only had a minor effect
791 on the type-II error rate in our simulation, the amount of items per factor had a
792 large effect on the type-II error rate. Importantly however for our findings, the
793 rejection rate of the one-step procedure was always equal to the rejection rate of
794 the MML based test. In fact, similar to in our main goodness-of-fit simulation,
795 AVI based tests are already very close to MML based test in terms of rejection
796 rates. It is still advisable to use the one step procedure, as this can provide the
797

⁷⁹⁸ theoretical guarantees discussed in the paper, as well as take the rejection rate
⁷⁹⁹ even closer to the MML based tests.